

UrbanStyle Database Audit: Week 2 Findings

Comprehensive Data Health, Relational Integrity, and Strategic Action Plan

Biggest Surprises & Key Highlights: Uncovering critical errors, unexpected data patterns, and high-impact actionable insights for strategic improvement.

Prepared by:	Team 6 — Executive Reporting
Target Audience:	Leadership & Cross-Functional Leads
Status:	Audited & Verified

Audit Scope & Navigation

Auditing four distinct database domains to isolate good practices, flag critical errors, and establish strategic business improvements.

1

Products Catalog

Assessing pricing logic, completeness, and duplication.

2

Customers Data

Evaluating field completeness and text-entry hygiene.

3

Sales Transactions

Purging duplicate bloat and validating revenue metrics.

4

Relational Integrity

Cross-validating orphan records and synchronization across all tables.

Domain 1: Products Data Health

Verified Good

100%

Critical Field Completeness

0 NULL names, 0 NULL prices, 0 NULL cost prices, 0 NULL categories.

100%

Logical Pricing

0 negative or extreme retail prices.

100%

Category Consistency

No inconsistent spellings or casing errors in classifications.

Attention Needed

12

Duplicate Names

12 product names appear more than once, affecting a total of 24 rows.

Actionable Insight: The products table is exceptionally clean. It only requires standard deduplication matching the methodology used for sales duplicates.

Domain 2: Customers Data Health

Good: 3,150 total rows with 0 NULL first/last names and 0 NULL phone numbers.

Error: 128 duplicate emails; 380 missing (NULL) emails.

Data Chaos Heat Map

54 total city entries exist for only 12 unique cities.

Tallinn (5 Variations)	" Tallinn "		
	"tallinn"		
	"Tallinn"		
	"TALLINN"		
	"Tallinn "		
Tartu	" Tartu "		
	"tartu"		
	"TARTU"		
Pärnu	" Pärnu "		
	"pärgu"		
	"PÄRNU"		

Text-entry hygiene is poor. Standardization of categorical text fields is required to ensure accurate geographic targeting.

Domain 3: Sales Transaction Surgery

Operations performed safely on staging environment (sales_test)

Total Database Rows	Raw 15,234	→	Cleaned 10,118	Net Change: -5,116 rows
Duplicate Invoices	Raw 5,116	→	Cleaned 0	Status: Purged
Impossible Future Dates	Raw 0	→	Cleaned 0	Status: Verified OK

Status Alert: Gross sales metrics are now clean and trustworthy. We can officially report exactly 10,118 unique orders to the Board of Directors.

Explanatory Insight: The Shopping Cart Effect

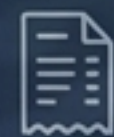
THE MYSTERY

Why did missing customer IDs drop from 1,487 to 988 after deduplication? Was data lost?

Multiple raw line items for an anonymous guest.



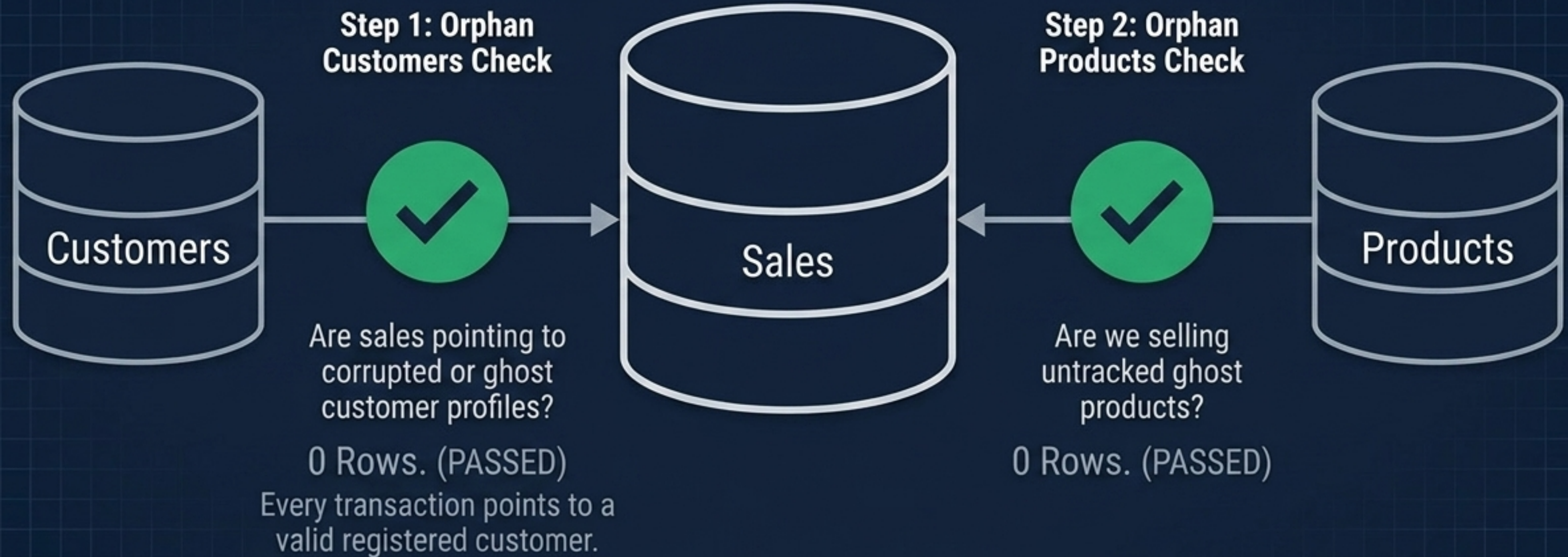
Line items share the same invoice_id, resulting in one consolidated receipt.



INVOICE_ID: INV-XXXX-YYYY
TOTAL ITEMS: 4 | TOTAL: \$XXX.XX

THE RESULT: Data was mathematically preserved, not lost. Post-cleanup, the multi-line carts consolidated into exactly **988 unique invoices**.

Domain 4: Relational Integrity & Cross-Validation



Conclusion: Structural integrity is pristine. The inventory catalog, customer profiles, and sales ledger are perfectly synchronized.

Business Assets: The Inefficiency Audit

Attention Needed

592

Dormant Customers

18.79% of the 3,150 registered customers have zero transactional history. A massive pool of acquired, inactive leads.

Monitor

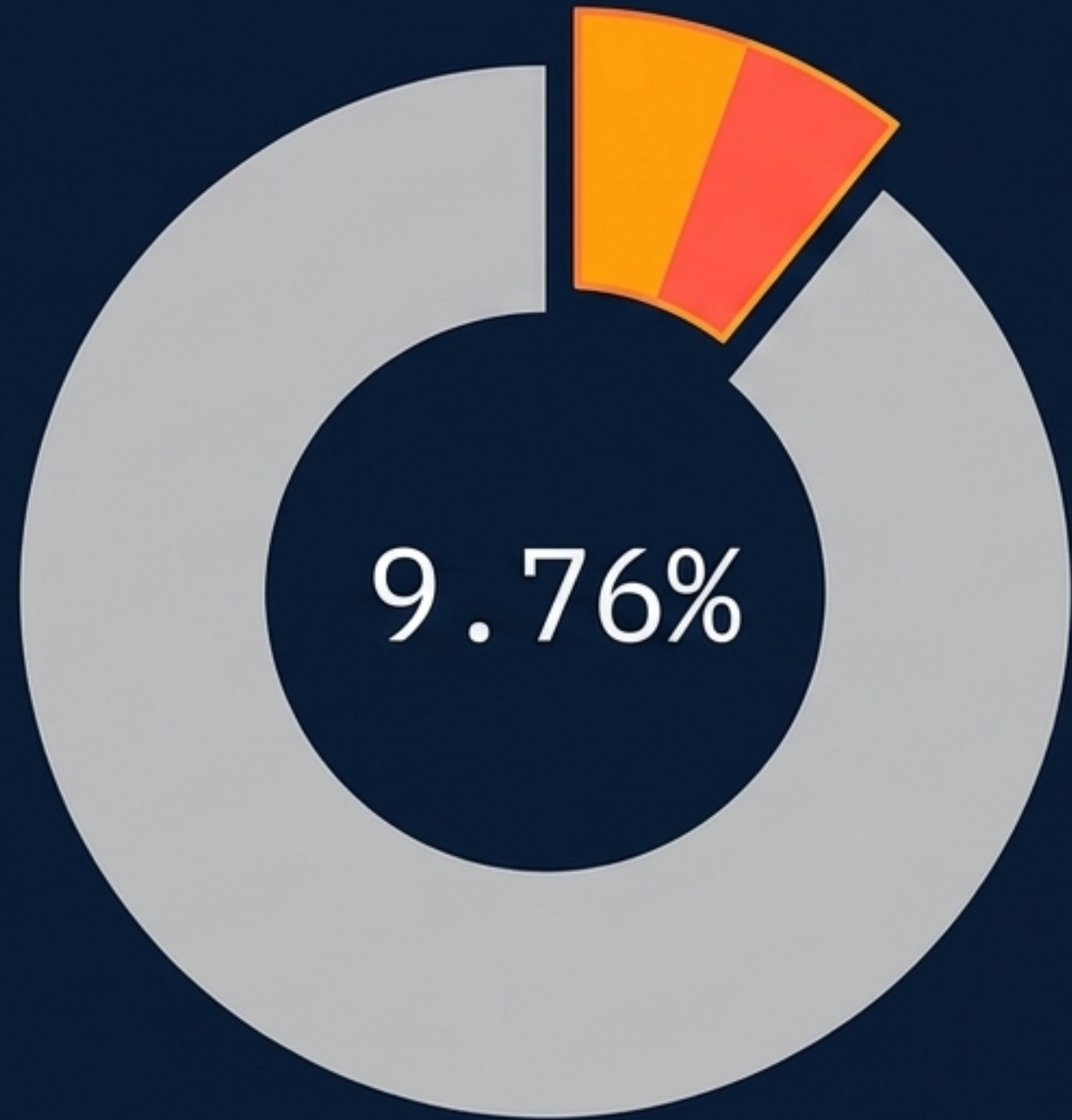
12

Dead Stock Products

12 products in the catalog have recorded exactly zero sales, tying up working capital and warehouse space.

The Insight: The database is technically verified but commercially inefficient. Action is required to activate dormant leads and clear unsold inventory.

The Strategic Blind Spot



The Finding

988 unique invoices belong to anonymous guest checkouts.

Exactly 9.76% of all transactions (988 of 10,118) are a complete marketing blind spot.

The Threat

While guest checkout prevents cart abandonment, Marketing Lead Anna Mets cannot retarget these users, calculate their Customer Lifetime Value (LTV), or include them in loyalty outreach.

Executive Summary & Diagnostic Conclusion

Structural Integrity

Excellent

Zero orphan records. Complete critical fields across products. No impossible dates or negative pricing. Relational links are perfectly sound.

Data Hygiene

Moderate

Significant duplicate transaction bloating (5,116 rows successfully resolved in staging) and severe inconsistencies in manual text entry (e.g., city names).

Business Utilization

Inefficient

Nearly 20% of registered leads are dormant, ~10% of total revenue is anonymous, and dead stock is occupying physical inventory space.

The Action Plan: **Prioritized Improvements**

IT & Architecture

(Toomas Kask)

1. Push the deduplicated sales_test cleaning script to **live production**.

2. Programmatically enforce **Foreign Key (FK)** & **UNIQUE** constraints to prevent future sync duplication and **orphan records**.

Marketing & Leadership

(Anna Mets & Kristi Tamm)

3. Launch targeted re-engagement campaigns (e.g., first-purchase discounts) for the **592 dormant customer IDs**.

4. Introduce registration incentives (e.g., 5% off) at checkout to convert the **9.76% anonymous guest segment** into **trackable leads**.

Operations & Inventory

(Marko Saar)

5. Conduct an immediate audit of the **12 unsold items** to optimize pricing, review demand, or free up physical warehouse space.